LAPORAN TUGAS 1 PEMROGRAMAN PARALEL



Dosen Pengampu:

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Source Code:

```
#include <stdio.h>
#include <chrono>
#include <iostream>
#include <omp.h>

double compute_pi_serial(long num_steps)
{
    double step = 1.0 / num_steps;
    double sum = 0.0;
    for (long i = 0; i < num_steps; i++) {
        double x = (i + 0.5) * step;
        sum += 4.0 / (1.0 + x * x);
    }
    return sum * step;
}

double compute_pi_parallel(long num_steps)
{
    double step = 1.0 / num_steps;
    double sum = 0.0;
    #pragma omp parallel
    {

    #pragma omp parallel for
    for (long i = 0; i < num_steps; i++) {
        double x = (i + 0.5) * step;
        sum += 4.0 / (1.0 + x * x);
    }
    }
    return sum * step;
}

int main()
{
    std::cout << "---------------------------------" <<
std::endl;
    std::cout << "NAMA : Nayunda Krisna" << std::endl;
    std::cout << "NPM  : 4524210077" << std::endl;
    std::cout << "---------------------------------" <<
std::endl;
```

```
    const long num_steps = 1000000000L;

    long m = omp_get_max_threads();
    std::cout << "Number of threads\t" << m << ".\n";

    auto start_time
        = std::chrono::high_resolution_clock::now();
    double pi_serial = compute_pi_serial(num_steps);
    auto end_time
        = std::chrono::high_resolution_clock::now();
    std::chrono::duration<double> serial_duration
        = end_time - start_time;

    start_time = std::chrono::high_resolution_clock::now();
    double pi_parallel = compute_pi_parallel(num_steps);
    end_time = std::chrono::high_resolution_clock::now();
    std::chrono::duration<double> parallel_duration
        = end_time - start_time;

    std::cout << "Serial result: " << pi_serial
            << std::endl;
    std::cout << "Parallel result: " << pi_parallel
            << std::endl;
    std::cout << "Serial duration: "
            << serial_duration.count() << " seconds"
            << std::endl;
    std::cout << "Parallel duration: "
            << parallel_duration.count() << " seconds"
            << std::endl;
    std::cout << "Speedup: "
            << serial_duration.count()
                    / parallel_duration.count()
            << std::endl;

    return 0;
}
```

Hasil Compile

```
C:\Users\Nayunda Krisna\Documents\1 UniVerse\Semester 5\Pemrograman Paralel\1>g++ -fopenmp OMP-PI-2.cpp -o OMP-PI-2

C:\Users\Nayunda Krisna\Documents\1 UniVerse\Semester 5\Pemrograman Paralel\1>OMP-PI-2.exe
---------------------------------
NAMA : Nayunda Krisna
NPM  : 4524210077
---------------------------------
Number of threads       16.
```